

układ mechaniczny opisany:

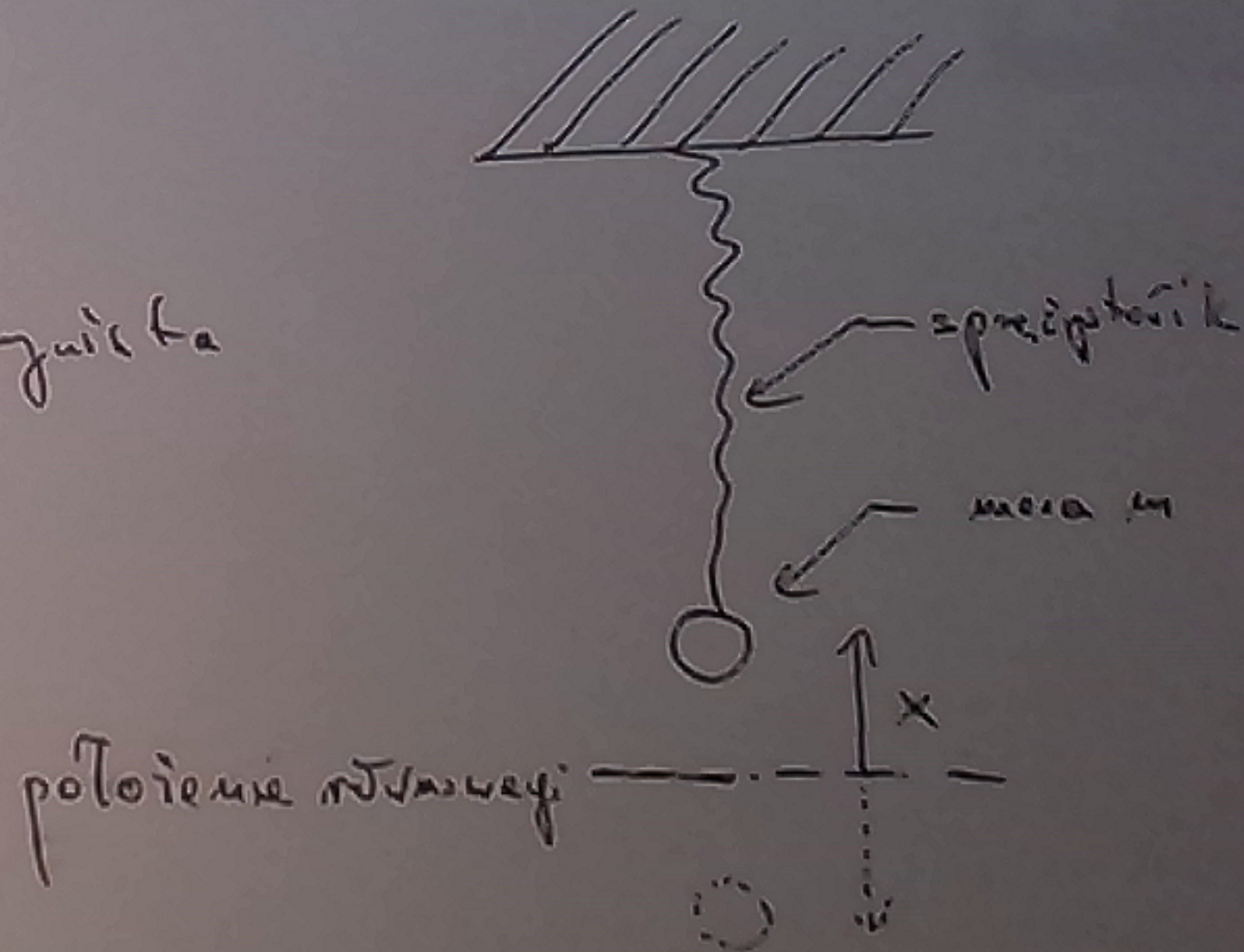
$x(t)$
↑
liczba mechaniczna

Dla uproszczenia:

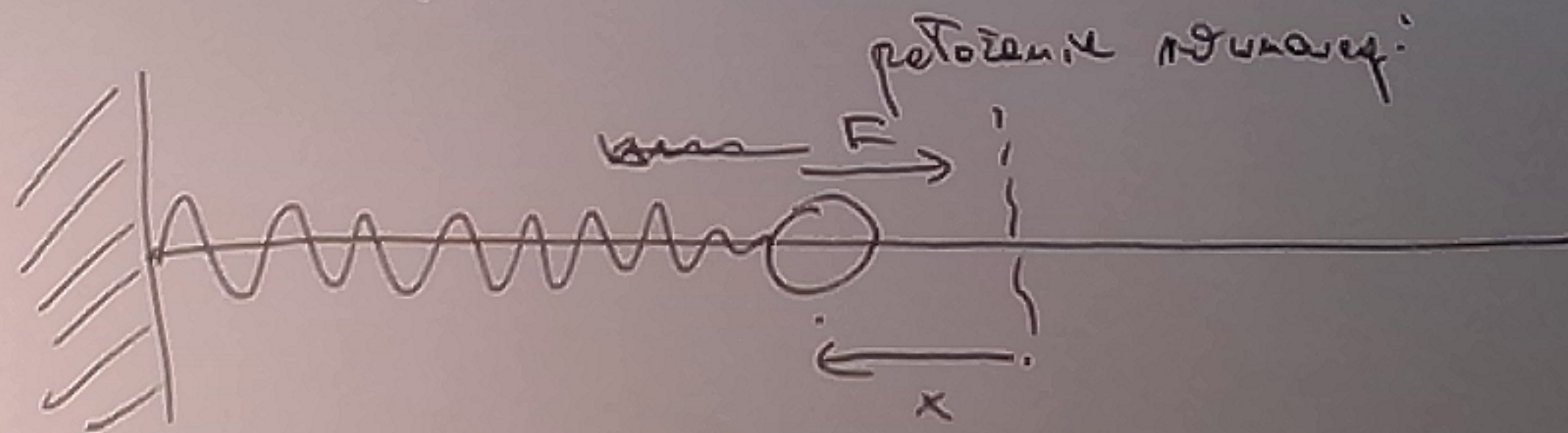
$$m = 1$$

$$k = 1$$

jednostki dowolne



Przebieg siły zastawionej przy tym układzie jest
płynący:

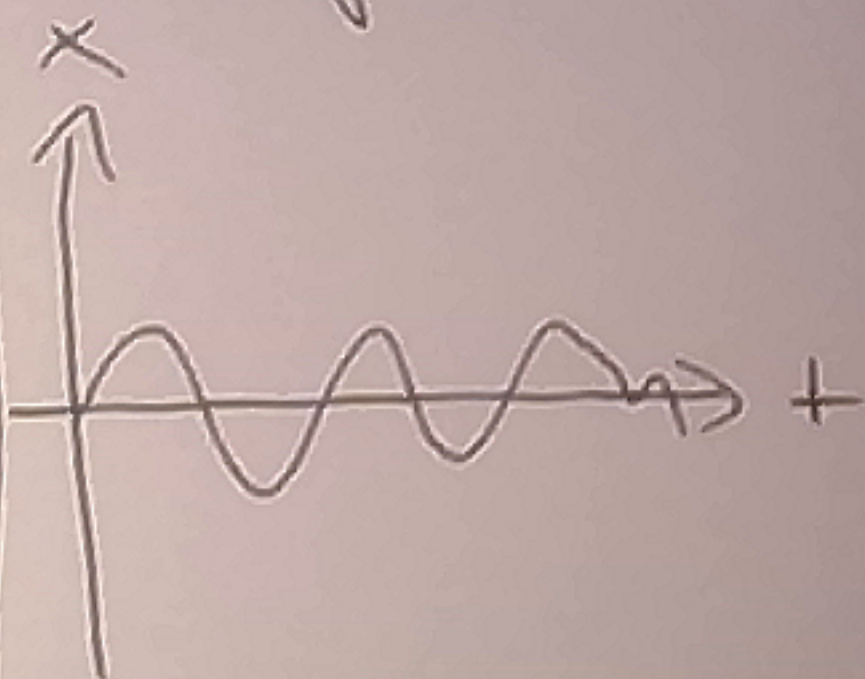


$$F = -kx$$

↑ wartość k jest sprężystości

$$S[x] = \int_{t_1}^{t_2} L(x(t), \dot{x}(t)) dt$$

kinetyczna



"działanie" "action"

"małe" zaburzenie

$$x \rightarrow x + \delta x$$

funkcja czasu

$$S[x + \delta x] = \int_{t_1}^{t_2} L(x + \delta x, \dot{x} + \delta \dot{x}) dt$$

$$\dot{x} = \frac{d}{dt}(x) \rightarrow \frac{d}{dt}(x + \delta x) = \dot{x} + \delta \dot{x}$$

$$= \int_{t_1}^{t_2} \left(L(x, \dot{x}) + \frac{\partial L(x, \dot{x})}{\partial x} \cdot \delta x + \frac{\partial L(x, \dot{x})}{\partial \dot{x}} \delta \dot{x} \right) dt =$$

$$= S[x] + \int_{t_1}^{t_2} \left(\frac{\partial L(x, \dot{x})}{\partial x} \cdot \delta x + \frac{\partial L(x, \dot{x})}{\partial \dot{x}} \delta \dot{x} \right) dt =$$

$$\int (\dot{f} g) dt = \int (\dot{f} g + f \dot{g}) dt$$

$$\int f \dot{g} dt = \int (\dot{f} g) dt - \int \dot{f} g dt$$

$$\frac{\partial L(x, \dot{x})}{\partial \dot{x}} \leftarrow \int \dot{f} g dt$$

$$= S[x] + \int_{t_1}^{t_2} \frac{\partial L(x, \dot{x})}{\partial x} \cdot \delta x dt + \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L(x, \dot{x})}{\partial \dot{x}} \delta x \right) dt - \dots$$

$$= \int_{t_1}^{t_2} \left(L(x, \dot{x}) + \frac{\partial L(x, \dot{x})}{\partial x} \cdot \delta x + \frac{\partial L(x, \dot{x})}{\partial \dot{x}} \delta \dot{x} \right) dt =$$

$$= S[x] + \int_{t_1}^{t_2} \left(\frac{\partial L(x, \dot{x})}{\partial x} \cdot \delta x + \frac{\partial L(x, \dot{x})}{\partial \dot{x}} \delta \dot{x} \right) dt =$$

$$\int (\dot{f} g) dt = \int (\dot{f} g + f \dot{g}) dt$$

$$\int f \dot{g} dt = \int (\dot{f} g) dt - \int f g \dot{g} dt$$

$$\frac{\partial L(x, \dot{x})}{\partial \dot{x}} \leftarrow \int \dot{L} \delta x$$

$$= S[x] + \int_{t_1}^{t_2} \frac{\partial L(x, \dot{x})}{\partial x} \cdot \delta x dt + \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L(x, \dot{x})}{\partial \dot{x}} \delta x \right) dt \dots$$

$$= \left[\frac{\partial L}{\partial x} \right]_{t_1} + \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x dt$$

$$+ \left(\frac{\partial L(x, \dot{x})}{\partial \dot{x}}(t_2) \delta x(t_2) - \frac{\partial L(x, \dot{x})}{\partial \dot{x}}(t_1) \delta x(t_1) \right) =$$

wzgl. δx

$$= \delta \left[\int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x dt \right] =$$

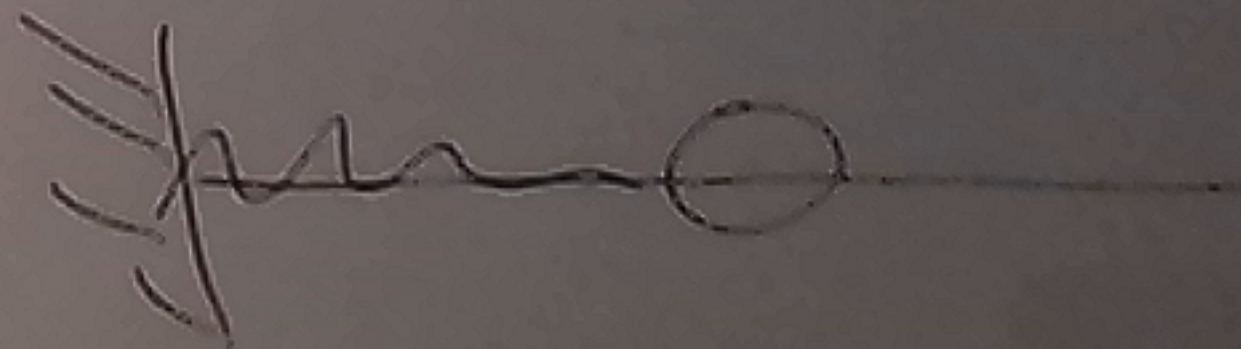
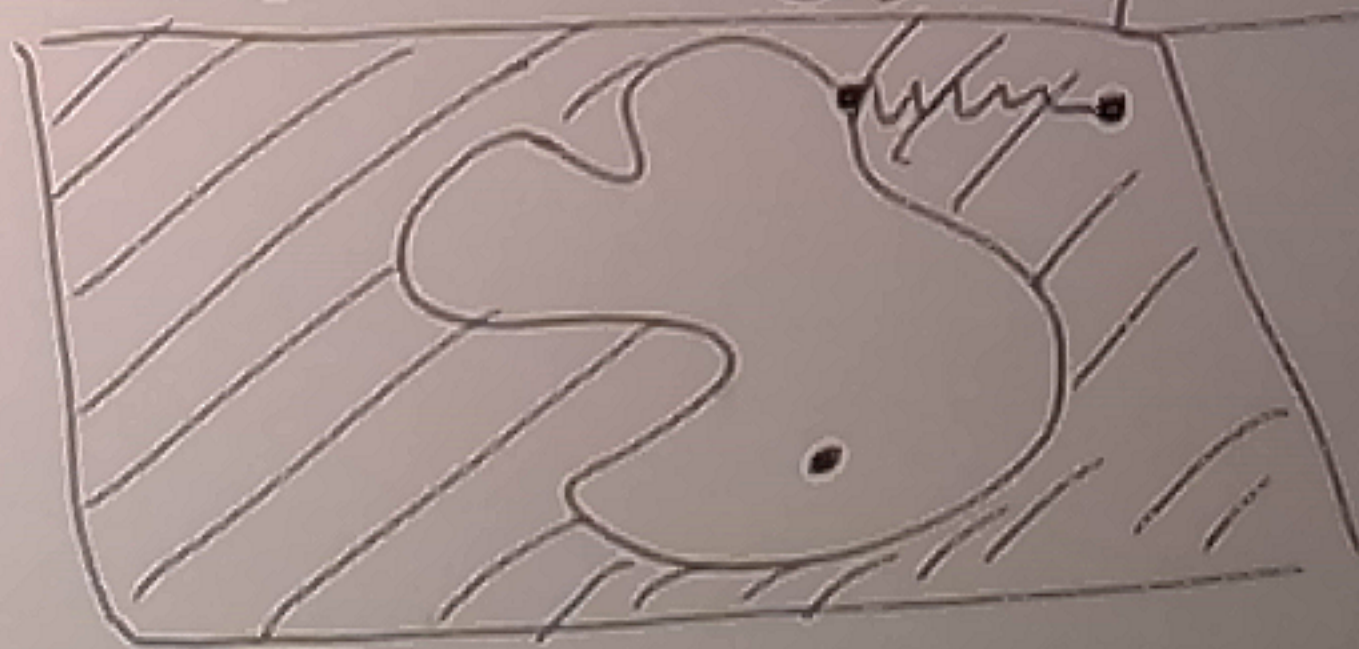
$$= \delta \left[\int_{t_1}^{t_2} \left(\frac{\partial L}{\partial x} - \frac{d}{dt} \frac{\partial L}{\partial \dot{x}} \right) \delta x dt \right]$$

Geželi trajektorija $x(t)$ spānā:

$$\frac{\partial L(x, \dot{x})}{\partial x} - \frac{d}{dt} \frac{\partial L(x, \dot{x})}{\partial \dot{x}} = 0$$

||

$$\delta S = 0 \quad \delta S = S[x + \delta] - S[x]$$

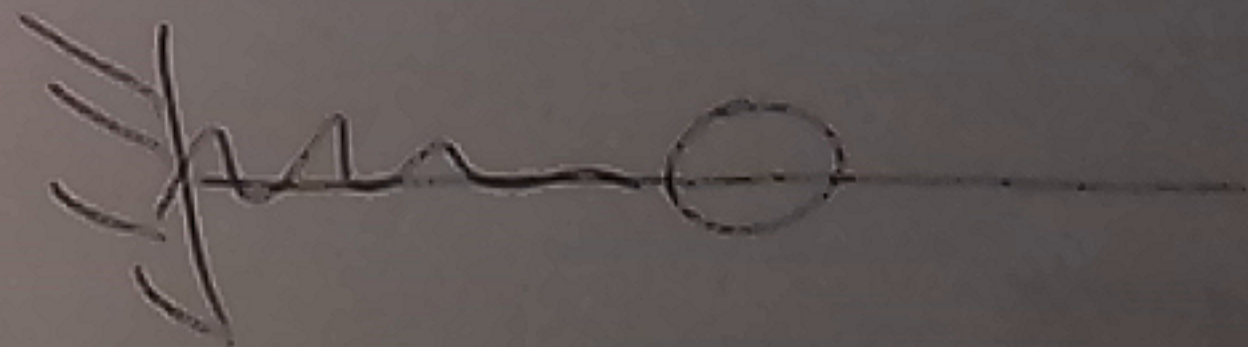
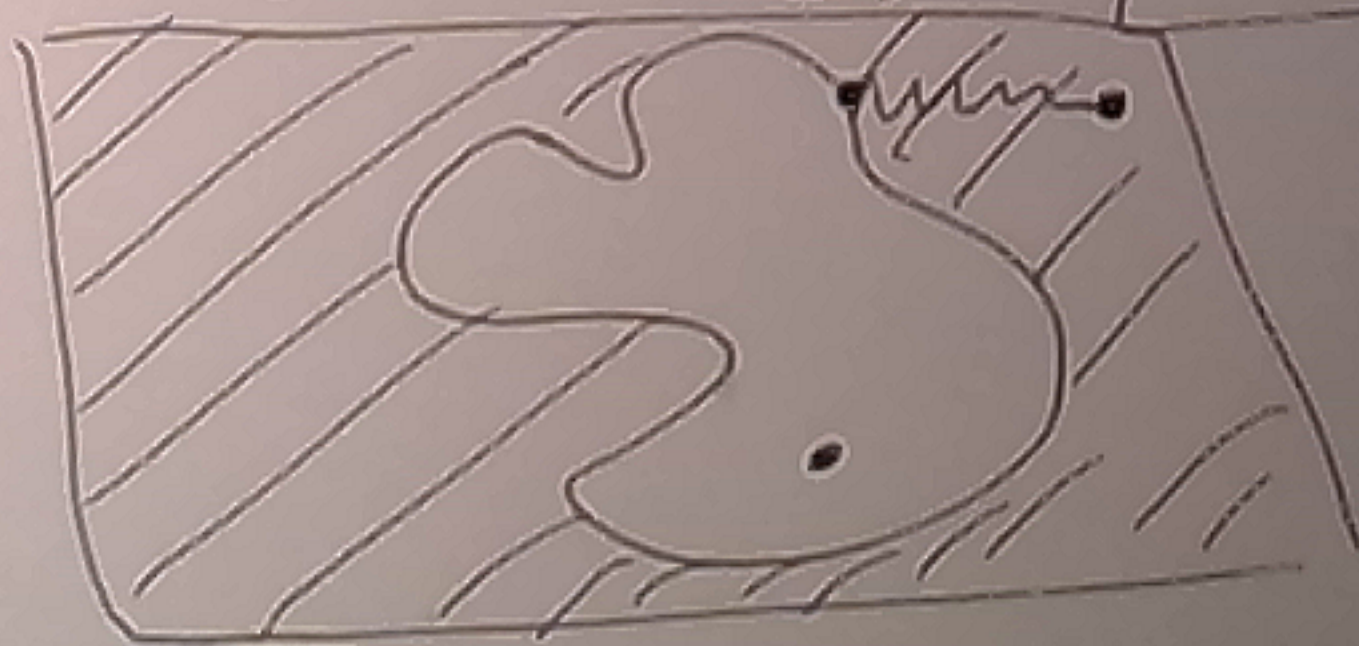


Meieli trajektoria $x(t)$ spētina:

$$\frac{\partial L(x, \dot{x})}{\partial x} - \frac{d}{dt} \frac{\partial L(x, \dot{x})}{\partial \dot{x}} = 0$$

$\Downarrow$

$$\delta S = 0 \quad \delta S = S[x + \delta] - S[x]$$



$$L = T - V$$

energie kinetisch $\frac{1}{2} k x^2$

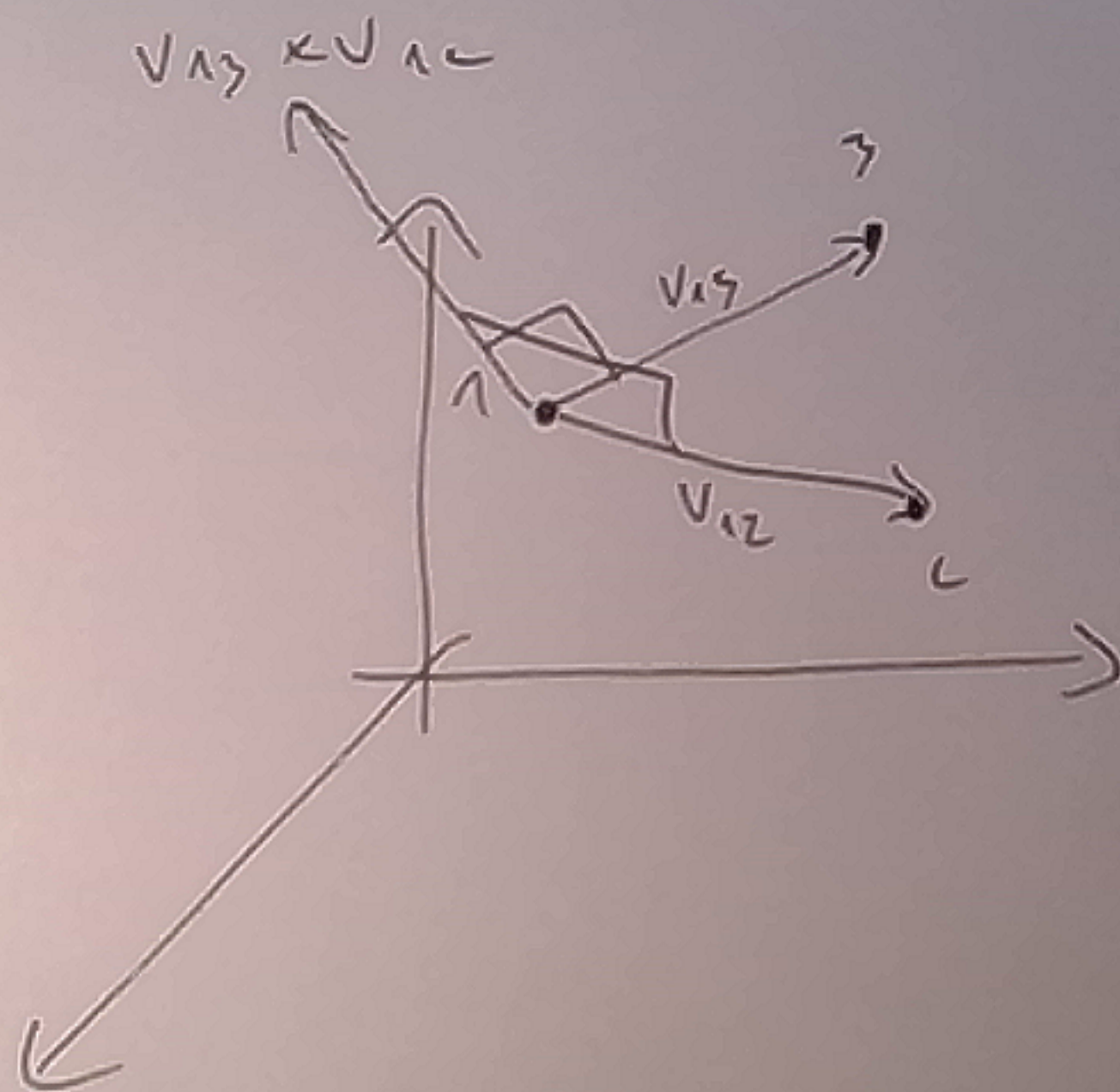
energie potenzial $\frac{1}{2} m \dot{x}^2$

$$\frac{1}{2} m \dot{x}^2$$

$$m = 1$$

$$k = 1$$

$$L = \frac{1}{2} \dot{x}^2 - \frac{1}{2} k x^2$$



$$\hat{m} = \frac{(v_{13} \times v_{12})}{|v_{13} \times v_{12}|}$$

$$|\hat{m}| = \underline{1}$$

